

Head Movement Biomarkers for PTSD, Depression, and Anxiety Disorders

Executive summary

The literature supports **head movement as a credible candidate digital marker**, but not yet a fully validated stand-alone clinical biomarker, for psychiatric assessment. The evidence is strongest in **major depressive disorder**: across clinically grounded interview studies, depressed groups often show **slower head motion, less positional change, longer downward or rightward dwell**, and reduced dynamic variability in **pitch/yaw/roll** trajectories. In head-only settings, reported discrimination is usually **moderate** rather than definitive, with performance around **71% average recall** in early severe-depression case-control work and about **79.6% mean accuracy** in a later remission-vs-depression interview study; multimodal fusion with face and voice consistently improves performance into the **mid/high 80s**. ¹

For **PTSD**, the evidence is much thinner and usually **multimodal rather than head-specific**. A 2013 ACII study reported that directly interpretable nonverbal indicators for depression/PTSD included affect, expression variability, and motor variability, and specifically highlighted **head-rotation variation** for PTSD with gender differences. A later trauma-survivor study using free-form interviews extracted movement features from video with **OpenFace** and reached **AUC 0.90 for PTSD** and **AUC 0.86 for MDD** at one month, but the paper did **not isolate axis-level head features** beyond broader movement/attentiveness/pupil-dilation variables. ²

For **anxiety**, the disorder-specific head-movement literature is sparser still. The clearest clinical result I found was not in primary anxiety disorder samples, but in **patients with severe depression**, where head-motion features predicted **psychological anxiety severity** with **MAE 0.31, R² 0.87**, and **class-level accuracy 0.75** after segmentation and stable feature selection. More direct social-anxiety video studies exist, but much of that evidence is currently **preprint-level**, often based on **self-report or task-induced anxious states** rather than clinician-confirmed DSM diagnoses. ³

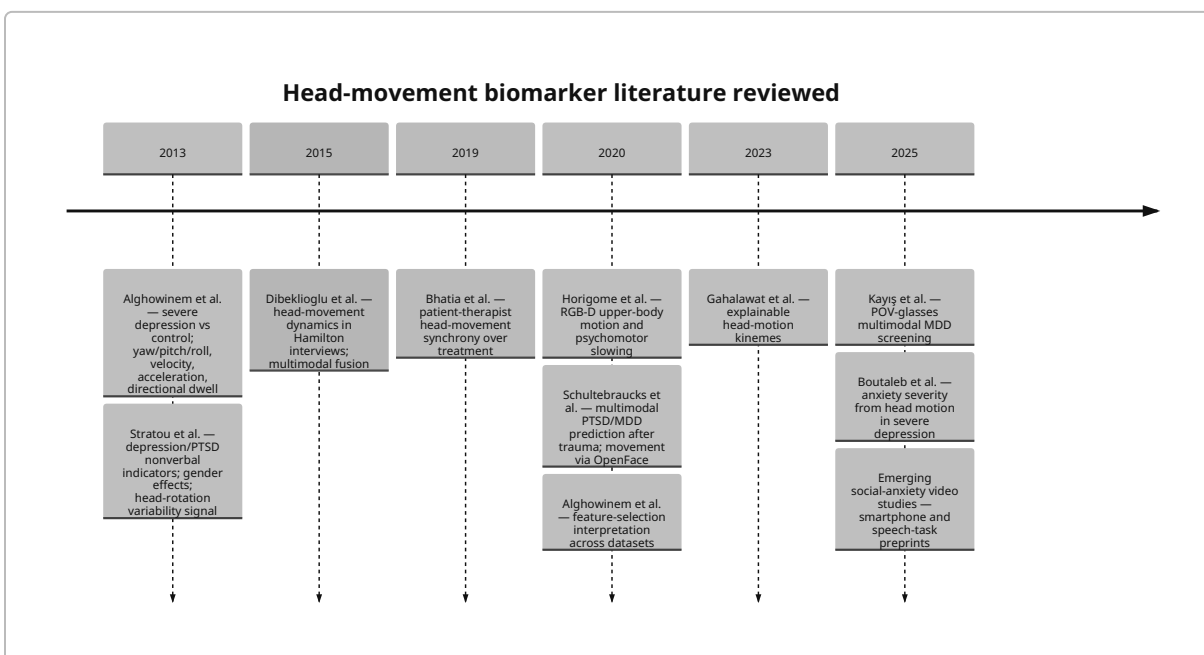
Methodologically, the field is still dominated by **offline pipelines**: framewise pose extraction, smoothing, hand-crafted statistical summaries, and retrospective model fitting. True online deployment is **technically feasible** with modern monocular toolkits such as **OpenFace** and **MediaPipe FaceMesh**, and the reviewed studies already operate at practical frame rates such as **25–30 fps**. But clinically serious work still depends heavily on controlled interviews, standardized distance/lighting, manual quality control, or post hoc segmentation. Robustness to **occlusion, diverse camera placement, masks/headgear, and real-world telehealth variability** is underreported. ⁴

One important scope note: the request refers to **two user-provided GitHub documents**, but those links were **not visible in the conversation context available to me**. I therefore could not directly review or merge their exact feature inventories and feasibility notes. The report below reconstructs the closest evidence-based feature inventory from the original papers and indicates where GitHub engineering notes would naturally map into the measurement pipeline.

Scope and study taxonomy

I prioritized **primary clinical studies** and original technical papers, with emphasis on papers traceable to **clinical interviews, clinically characterized samples, or validated depression/PTSD tasks**. Because disorder-specific head-movement evidence is concentrated in depression and is sparse for PTSD/anxiety, I also included a small number of **recent preprints** for anxiety when they were directly on-point and clearly labeled as such. Across the studies I could verify, the literature is heavily skewed toward **MDD/depression**, with PTSD represented mainly by **multimodal predictive studies** and anxiety represented by a mix of **comorbid-with-depression** and **social-anxiety screening** studies. ⁵

The study stream is easiest to understand as a progression from **simple axis-based pose statistics** to **dynamic trajectory modeling**, then to **interaction-level synchrony** and **higher-order pattern units** such as “kinemes.” The timeline below summarizes the studies reviewed here. ⁶



At the feature level, the literature converges on a compact inventory. The “classical” family includes **pitch, yaw, roll**, plus first and second derivatives (**velocity, acceleration**), and summary statistics such as **mean, variance, standard deviation, range, skewness, kurtosis**, and **durations of directional states**. More recent papers add **movement bouts, steady vs moving segmentation, rapid movement counts, synchrony**, and **kineme-like segment clusters**. Across studies, **pitch and yaw** repeatedly appear more informative than **roll**, and **dynamic variability** is often more useful than raw static pose alone. ⁷

Depression evidence

The clearest head-specific depression paper in the set is **Alghowinem et al. 2013**, which used a clinically validated case-control dataset of **30 severely depressed participants and 30 healthy controls** recorded during open-ended interview questions. The pipeline built a **subject-specific Active Appearance Model** using **68 landmarks**, matched it to a **58-point 3D face model** through **46 corresponding points**, and

estimated head pose with **POSIT** from **monocular 2D video**. From each frame at **25 fps**, the authors extracted **yaw, roll, and pitch**, then computed **velocity and acceleration**, creating **9 low-level framewise signals**, plus **100 statistical “functionals”** such as maxima/minima, ranges, variability, durations of looking left/right/up/down, direction-change rates, and counts of direction changes. They removed bad frames with **Grubbs’ test** and used **leave-one-subject-out** evaluation with **GMM+SVM** for low-level features and **SVM** for functionals. ⁸

That paper remains important because it did not just report classification—it also described the behavioral signature it found. The head-only system reached **68.8% average recall** with low-level features and **76.8%** with selected statistical functionals, averaging **71.2%** overall. The most interpretable differences were **slower head movements, less change of head position, longer duration looking right, and longer looking down**, which the authors linked to fatigue/psychomotor retardation and possible eye-contact avoidance. Several feature-level tests were significant, including **max pitch velocity** ($p = 0.01$), **max pitch acceleration** ($p = 0.02$), **range of pitch acceleration** ($p = 0.03$), **max duration of looking right** ($p = 0.02$), and **average duration of looking down** ($p = 0.05$, and $p = 0.01$ in the “good news” subset). ⁹

Dibeklioglu et al. 2015 shifted the field from static summaries to **motion dynamics in longitudinal treatment interviews**. Their sample comprised **48 participants** drawn from a clinical trial for depression treatment, with **95 sessions** usable after filtering; sessions with **HRSD > 14** were labeled depressed and **HRSD < 8** remitted. Video came from a **VGA camera** at roughly **15° off frontal** view. Facial shape and head pose were tracked with **ZFace**, giving **49 facial landmarks** and **three rigid head degrees of freedom**. Head trajectories for **pitch, yaw, roll** were **smoothed with 4253H-twice**, then converted to amplitude, velocity, and acceleration time series, and described with **21 dynamic features** per time series, including maxima, means, standard deviations, and positive/negative segment frequencies. Feature selection used **mRMR**, and classification used **logistic regression** with **leave-one-participant-out** validation. ¹⁰

The head-dynamics results were strong, though still clearly improved by multimodal fusion. Head-only dynamics achieved **79.59% mean accuracy** for remitted vs depressed sessions, outperforming earlier head-movement encodings from the same general line of work. Head and facial dynamics each beat vocal prosody, and all three together reached **88.93%**. The modality comparison itself was significant ($F_{2,282} = 3.16$, $p = 0.04$), and both facial and head movement dynamics outperformed prosody ($p = 0.02$ and $p = 0.03$, respectively). This is one of the clearest demonstrations that **head movement is useful, but usually best as one channel in a multimodal system rather than a solitary marker**. ¹¹

Horigome et al. 2020 approached the same psychomotor question from a different angle: **RGB-depth sensing** of the upper body during clinical interviews. Their sample included **47 participants** contributing **144 datasets: 16 healthy controls, 17 MDD, and 14 bipolar disorder** cases. The abstract reports that depressed patients moved **significantly slower** than controls at the **5th percentile** and **50th percentile** of motion speed, and that **HAMD-17** correlated with the **5th percentile, 50th percentile, and mean speed**. ML prediction of the presence and/or severity of depressive symptoms achieved only $\kappa = 0.37\text{--}0.43$, so the paper supports the biological relevance of slowed motion more than it supports near-clinical predictive accuracy. Importantly, this was an interview-based, clinically grounded study, but details such as the exact per-frame pose estimator and smoothing pipeline are not specified in the accessible abstract. ¹²

Bhatia et al. 2019 moved from single-person dynamics to **interpersonal head-movement synchrony** during depression interviews. The sample involved **49 patients with MDD**, observed at up to four visits over **21 weeks**, for **113 observations** in the final analysis. Interviews were recorded with **four**

synchronized cameras at 30 fps, and **ZFace** tracked **pitch, yaw, and roll** from 2D videos; the reported head-tracking success rate was **98.3%** for patients and **89.4%** for therapists. Synchrony between patient and therapist was computed using **windowed cross-correlation** and **peak-picking**, then analyzed with a **gamma-family hierarchical linear mixed model**. ¹³

The result is notable because it is a **negative finding** that still matters methodologically. For both pitch and yaw synchrony, **HRSD was not significant (pitch $p = 0.490$; yaw $p = 0.859$)**. The variance structure showed strong **dyad-level ICCs (0.81 for pitch, 0.865 for yaw)** and weak visit-level ICCs (**0.08** and **0.064**), meaning synchrony varied more within dyads over time than between dyads. The authors explicitly suggested that time-domain synchrony may have missed clinically meaningful changes and recommended **time-frequency analysis** as future work. This paper is one of the few to show that not every plausible head-movement marker survives rigorous longitudinal modeling. ¹⁴

The most explicit attempt to understand *which* head features matter comes from **Alghowinem et al. 2020**, a feature-selection interpretation study spanning **three real-world depression datasets** and **902 behavioral cues** from speech, eye activity, and head pose. The paper's headline result is that among head features, **yaw-head-movement features** emerged most consistently across datasets, especially **average duration of looking left** and **speed-related features** for yaw, with pitch and roll contributing mainly through movement-speed groups. The paper also emphasizes that small, carefully selected feature sets—**as few as 9 features**—could outperform using all available features, which is directly relevant to feasibility and reproducibility. ¹⁵

A more recent explainable direction appears in **Gahalawat et al. 2023**, which treated head motion as sequences of elementary units ("**kinemes**") extracted from **OpenFace 3D Euler rotations**. For the **BlackDog** dataset, the paper uses **60 subjects (30 depressed, 30 controls)**, and for **AVEC2013** it analyzes the challenge corpus. Videos were segmented into overlapping windows; the head-pose sequences were clustered with **GMM-based kineme discovery**, then summarized either by class-discriminative kineme frequencies or by reconstruction-error statistics relative to control-only kinemes. On BlackDog, the best reported performance was **chunk-level F1 0.79** and **video-level F1 0.80** with interpretable kineme-based descriptors; the paper also shows that **pitch** is the single most informative axis and **pitch + yaw** usually the best combination, while **roll** is the weakest. The **AVEC2013 video-level F1 of 1.00** is strikingly high and should be read cautiously until independently replicated, because such ceiling performance is unusual for psychiatric video classification. ¹⁶

Kayış et al. 2025 is clinically interesting because it used **clinician-worn POV glasses** rather than a fixed front camera. The sample included **44 MDD patients** and **41 age/sex-matched controls**, with diagnoses confirmed using **SCID-5** and MDD severity screened with **BDI**. The setup used **1080p video at 30 fps, 400–600 lux** lighting, and roughly **100 cm** interviewer-participant distance. Visual analysis used **OpenCV**, **MediaPipe FaceMesh** for 3D head pose from 2D video, **Py-Feat/FACS** for facial expression, and **Whisper** for speech. Head movement features were deliberately simple: **total head movement** as cumulative absolute angular change across **pitch/yaw/roll**, and **rapid head movements** as frame-to-frame changes exceeding **5°** on any axis. ¹⁷

What makes this paper so useful for interpretation is that it shows the current limits of simple head metrics. Neither **total head movement ($p = 0.418$)** nor **rapid head movement ($p = 0.843$)** differed significantly between MDD and controls, even though the overall multimodal classifier still performed well, reaching **84.7% accuracy, 90.9% sensitivity, 78% specificity**, and **F1 = 86%** with **leave-one-out** validation. In other

words, the paper supports head movement as a **complementary feature family**, but it also suggests that more nuanced dynamics may matter more than coarse totals or binary “rapid movement” counts. ¹⁸

PTSD and anxiety evidence

The earliest clearly relevant PTSD paper I found is **Stratou et al. 2013**, which introduced a **semi-structured virtual human interview dataset with 53 recorded interactions** and framed depression/PTSD prediction using **directly interpretable nonverbal indicators** drawn from **affect**, **expression variability**, and **motor variability**. The accessible sources do not expose the full methods table or performance metrics, but they do show two clinically relevant points: first, the authors argue that **gender materially affects automatic assessment** of both depression and PTSD; second, their figure summary highlights **head-rotation variation** as a meaningful PTSD-related signal. Because the accessible sources do not provide the exact tracker, frame rate, or reported AUC/accuracy values, those details have to remain marked as **unspecified in the accessible version**. ¹⁹

The stronger modern PTSD evidence comes from **Schultebrucks et al. 2020**, a multimodal “clinical signs” study in **81 trauma survivors** assessed after emergency-department admission. The study extracted **247 features** spanning face, voice, speech content, and movement. Movement variables were extracted from raw MP4 video using **OpenFace in Python**, and the movement domain included **head movement**, **attentiveness**, and **pupil dilation rate**. Data preprocessing used the **caret** package, **one-hot encoding**, normalization to **[0,1]**, near-zero-variance filtering, and **k-nearest-neighbor imputation** for at most **1%** missingness; models were deep neural networks with both internal test and nested cross-validation evaluation. ²⁰

The predictive performance was strong: the internal test set achieved **AUC 0.90** for PTSD status with **precision 0.83, recall 0.84, F1 0.83**, and **AUC 0.86** for MDD with **precision 0.83, recall 0.82, F1 0.82**. In nested cross-validation, PTSD remained at **AUC 0.90** with **weighted precision 0.90, recall 0.89, F1 0.90**, and MDD reached **AUC 0.88**. The paper states that all four domains ranked highly among the top features, but the movement domain was **not decomposed into axis-level head kinematics** in the main text. That makes this one of the strongest *predictive* PTSD papers in the review, but only a **moderately informative head-biomarker paper** in the narrower feature-engineering sense. ²¹

For **anxiety**, the most directly useful clinical head-movement paper is **Boutaleb et al. 2025**, which used the **CALYPSO** depression dataset to predict **psychological anxiety** in **32 patients with severe depression**. The interviews included an informal conversational segment and a later structured HDRS assessment; the anxiety target was the psychiatrist-rated **psychological anxiety score from 0 to 4**. The paper automatically extracted head pose, speed, and acceleration from interview videos, then used a motion-segmentation strategy to separate **moving vs steady periods**. Statistical descriptors from global and segmented sequences yielded a **283-feature vector**, which was filtered with **sequential feature selection** and modeled with **Lasso regression** under **10-fold cross-validation**. ²²

The best model used only **14 features** and achieved **MAE 0.31, R² 0.87**, and discretized **accuracy 0.75**. A baseline without the GMM-style motion segmentation deteriorated badly to **MAE 0.90, R² 0.10**, and **accuracy 0.44**, which is unusually direct evidence that **how head trajectories are segmented matters at least as much as which summary statistics are used**. The most interpretable features were **median steady pitch**, **temporal skewness during moving segments**, and **visits per minute** between moving and steady states, with the qualitative conclusion that **higher anxiety corresponded to longer, more irregular**

motion bouts and fewer movement-state transitions. The same approach worked less well for **somatic anxiety** (MAE 0.47, R^2 0.53), implying that head motion may be more closely tied to **psychological/cognitive anxiety** than to body-somatic tension alone. ²²

Beyond that, the anxiety literature becomes much less clinically mature. **Beyond Questionnaires: Video Analysis for Social Anxiety Detection** reports **92 participants** performing an impromptu speech task and uses **head pose, body pose, eye gaze, and action units** from video to classify social anxiety with up to **74% accuracy**; however, the paper's labels are based on **questionnaire-derived categorization in a stress task**, not clinician-confirmed DSM diagnosis. That makes it useful as **proof of signal**, but not yet as evidence of a validated diagnostic biomarker. ²³

Similarly, **AnxietyFaceTrack** reports **91 participants** recorded by **low-cost smartphone cameras** in an unstaged social setting. It used **OpenFace** to extract **714 facial features**, including **head location and rotation**, and found that head-pose features alone achieved about **85% multiclass accuracy** while post-hoc analysis repeatedly highlighted **pose.Rx / head pitch rotation** among the most important features. Overall performance rose to **91%** multiclass accuracy and around **92.3% average** across binary classifications when the top 20% of features were used with a **Random Forest**. Again, though, this is currently a **preprint/social-state screening study**, not a clinician-confirmed anxiety-disorder biomarker paper. ²⁴

Taken together, the PTSD/anxiety evidence supports a cautious conclusion. **Head movement probably contains clinically meaningful information in PTSD and anxiety**, but the available evidence does **not yet isolate a stable, axis-level, disorder-specific head signature** the way the depression literature begins to do. For PTSD, head movement is mostly embedded inside multimodal models. For anxiety, the best direct result is anxiety-severity regression **within severe depression**, while social-anxiety papers remain closer to **screening/proof-of-concept** than to biomarker validation. ²⁵

Measurement pipelines and feasibility

Across the reviewed papers, four measurement paradigms recur. The oldest is **monocular 2D video plus a fitted 3D face model**, as in Alghowinem's **AAM + POSIT** workflow. A second is **person-independent 3D face alignment from 2D video**, as in **ZFace**. A third is **landmark-plus-face-mesh monocular estimation** using modern consumer hardware, as in **MediaPipe FaceMesh**. The fourth is **depth-assisted tracking** with **RGB-D sensors**. Each choice trades off accuracy, calibration burden, and deployability. ²⁶

The older AAM/POSIT workflow extracted detailed axis-level kinematics, but it also reveals why many papers remained offline. Alghowinem's system used **subject-specific AAMs**, required **semi-automated initialization**, sometimes needed **manual correction when fitting was inaccurate**, and then removed bad frames statistically. That is scientifically careful, but it is not a low-friction production workflow. It also suggests sensitivity to face-detection failure, partial occlusion, and non-frontal movement—exactly the conditions common in telehealth or home monitoring. ²⁷

ZFace-based work looks more scalable. In Dibeklioglu 2015 it enabled person-independent 3D registration from **single-camera 2D video**; in Bhatia 2019 it supported synchrony analysis on **30 fps multi-camera** interview recordings, with tracking success of **98.3%** for patients and **89.4%** for therapists. Those asymmetrical success rates are useful: even in a controlled clinical setting, camera angle and role-specific

posture are enough to reduce tracker yield. That is a concrete warning for real-world deployment—**therapist-view, side-view, or occluded profiles can materially lower usable frame counts.** ²⁸

The newest clinically deployed pipeline in the set is the **POV glasses** study, which used **1080p/30 fps, controlled lighting of 400–600 lux, fixed face-to-face distance**, and **MediaPipe FaceMesh** for head pose from 2D video. This setup is comparatively favorable for operational use: monocular, consumer-grade, no patient-worn device, and compatible with real clinical flow. At the same time, the paper’s strong environmental standardization implies that the reported results are **not yet evidence of robustness to uncontrolled lighting, backlighting, camera shake, motion blur, or variable speaking distance.** ²⁹

That leads to the central feasibility conclusion. **Real-time head-pose extraction is feasible; real-time psychiatric inference is much less validated.** The feature extraction part now fits ordinary hardware and frame rates. The inference part in the literature still usually depends on **smoothing, PCA or clustering, cross-validated feature selection, state segmentation, or aggregation over long interview windows.** The Gahalawat study shows that performance improves when multiple “thin slices” are aggregated, especially on longer **75–105 s** chunks in BlackDog, while the Boutaleb study shows segmentation is critical to prediction quality. Those are signs of a field whose *signal extraction* is becoming easy but whose *clinical modeling* is still fundamentally offline. ³⁰

In practice, the most robust feature families for online monitoring are probably the simplest ones: **pitch/yaw/roll**, their **smoothed first derivatives, movement-bout durations**, and **rapid-movement counts**. These features are cheap to compute and interpretable. More ambitious constructs such as **kinemes, interpersonal synchrony**, or **segment-level skewness/kurtosis** are valuable, but they require either larger temporal context or more careful drift control. The reviewed literature also suggests that **pitch and yaw should be prioritized over roll**, because roll was repeatedly weaker in explainable analyses. ³¹

A pragmatic engineering stack, based on what the reviewed papers actually use, would look like this: capture **30 fps RGB video**; estimate **3D head pose** via **MediaPipe** or **OpenFace**; smooth each axis with a short robust filter; compute axis amplitudes, velocities, accelerations, movement-state transitions, and directional dwell; attach per-frame or per-window **tracking confidence**; and only then run a psychiatric model. The code fragment below is deliberately simple, but it reproduces the core logic of several reviewed pipelines. The surrounding feature choices are consistent with the OpenFace- and MediaPipe-based studies cited above. ³²

```
import numpy as np
import pandas as pd

# Example input: OpenFace-style CSV with pose_Rx, pose_Ry, pose_Rz in radians
df = pd.read_csv("openface_output.csv")

# Fill short gaps and smooth each pose axis
fps = 30.0
dt = 1.0 / fps
axes = ["pose_Rx", "pose_Ry", "pose_Rz"] # pitch, yaw, roll in many OpenFace exports
feat = {}
```

```

for ax in axes:
    x = (
        df[ax]
        .interpolate(limit_direction="both")
        .rolling(window=5, center=True, min_periods=1)
        .median()
        .to_numpy()
    )
    v = np.gradient(x, dt)
    a = np.gradient(v, dt)

    moving = np.abs(v) > np.percentile(np.abs(v), 60) # simple adaptive
threshold
    transitions = np.sum(np.diff(moving.astype(int)) != 0)

    feat[f"{ax}_mean"] = float(np.mean(x))
    feat[f"{ax}_std"] = float(np.std(x))
    feat[f"{ax}_vel_mean_abs"] = float(np.mean(np.abs(v)))
    feat[f"{ax}_acc_mean_abs"] = float(np.mean(np.abs(a)))
    feat[f"{ax}_moving_fraction"] = float(np.mean(moving))
    feat[f"{ax}_transitions_per_min"] = float(transitions / (len(x) * dt /
60.0))

# Example summary feature vector
feature_vector = pd.Series(feat)
print(feature_vector.sort_index())

```

Comparative matrix and disorder-level interpretation

The table below condenses the most informative studies I found into a single matrix. When a paper did not specify an item in the accessible source, I label it **not specified** rather than infer it.

Study	Population and disorder	Head features	Measurement and preprocessing	Statistical / ML method	Performance and limitations
Alghowinem et al. 2013 ⁸	30 severe depression, 30 healthy controls; depression	Per-frame yaw, pitch, roll , plus velocity and acceleration ; directional dwell, direction-change counts, steady/continuous movement durations	Subject-specific AAM with 68 landmarks; 46 matched to 3D model; POSIT; 25 fps ; outlier removal by Grubbs' test ; functional summaries	GMM+SVM for low-level features, SVM for functionals; leave-one-subject-out	Head-only average recall around 71.2% ; strongest selected-functionals 76.8% AR . Significant signals included slower pitch dynamics and longer down/right dwell. Limitation: small sample, manual fitting burden, controlled interviews.
Dibeklioglu et al. 2015 ³³	48 treated depression patients, 95 sessions; remitted vs depressed by HRSD	Pitch, yaw, roll amplitudes, velocities, accelerations; 21 dynamic features per axis	ZFace person-independent 3D registration from 2D video; 4253H-twice smoothing ; dynamic segmentation into increasing/decreasing segments; mRMR	Logistic regression , leave-one-participant-out	Head-only 79.59% mean accuracy ; face+head 85.37% ; head+face+voice 88.93% . Limitation: no healthy controls; exclusion of intermediate HRSD scores narrows ecological validity.

Study	Population and disorder	Head features	Measurement and preprocessing	Statistical / ML method	Performance and limitations
Bhatia et al. 2019 13	49 MDD patients, 113 observations across treatment; depression severity tracking	Pitch/yaw synchrony between patient and therapist; roll tracked but not main synchrony target	ZFace on synchronized 30 fps multi-camera videos; windowed cross-correlation, peak picking	Hierarchical linear mixed model	Negative result for severity: HRSD $p = 0.490$ for pitch synchrony, $p = 0.859$ for yaw synchrony. Strong dyad ICCs (0.81–0.865). Limitation: therapist rotation across sessions; time-domain synchrony may miss frequency-structured effects.
Horigome et al. 2020 12	16 HC, 17 MDD, 14 bipolar; 144 datasets; depression symptoms	Interview head motion speed summaries; 5th percentile, 50th percentile, mean speed	RGB-D sensor in clinical interview; exact smoothing/calibration not specified in accessible abstract	Group comparison plus ML for severity presence/level	Depressed groups moved slower at 5th and 50th speed percentiles; HAMD-17 correlated with motion-speed summaries; ML only $\kappa = 0.37–0.43$. Limitation: small sample, especially severe/young cases.

Study	Population and disorder	Head features	Measurement and preprocessing	Statistical / ML method	Performance and limitations
SchulteBraucks et al. 2020 ²⁰	81 trauma survivors after ED admission; PTSD and MDD	Movement domain included head movement, attentiveness, pupil dilation rate ; exact axis-level head variables not specified	Raw MP4 video processed with OpenFace (Python) ; normalization, near-zero-variance removal, kNN imputation	Deep neural networks + nested cross-validation	PTSD AUC 0.90, MDD AUC 0.86 in internal testing; nested CV AUC 0.90 PTSD, 0.88 MDD . Limitation: head features embedded within multimodal model; head-only contribution not isolated.
Alghowinem et al. 2020 ³⁴	Three depression datasets combined; depression	902 behavioral cues overall; yaw-head-movement feature groups most consistently selected; pitch/roll contributed mainly via speed groups	Multi-dataset feature extraction; 38 feature-selection algorithms ; robust aggregation framework	Multiple classifiers across selected subsets	Small selected sets—down to 9 features —outperformed all-features models. Limitation: interpretability study rather than one clean clinical cohort; performance depends on dataset harmonization.

Study	Population and disorder	Head features	Measurement and preprocessing	Statistical / ML method	Performance and limitations
Gahalawat et al. 2023 ³⁵	BlackDog: 30 depressed + 30 controls; AVEC2013 challenge corpus; depression	OpenFace pitch/yaw/roll; overlapping segment clusters (" kinemes "); reconstruction-error statistics	OpenFace 3D Euler angles; overlapping windows; GMM kineme discovery; chunk and video aggregation	LR, RF, SVC, XGB, MLP	Best on BlackDog: chunk F1 0.79, video F1 0.80 ; AVEC video-level scores reached 1.00 in some settings. Pitch and pitch+yaw were strongest; roll weakest. Limitation: challenge-dataset generalization and unusually high AVEC results warrant replication.
Kayış et al. 2025 ³⁶	44 MDD, 41 controls; MDD detection	Total head movement and rapid head movement (>5° frame-to-frame across pitch/yaw/roll)	Clinician-worn POV glasses, 1080p/30 fps, MediaPipe FaceMesh, OpenCV, standardized lighting/distance	Multiple classifiers; ExtraTrees primary; LOOCV	Overall multimodal classifier: 84.7% accuracy, 90.9% sensitivity, 78% specificity, F1 86% . Head metrics alone were not significant (p = 0.418, 0.843) . Limitation: simple head features may be too coarse.

Study	Population and disorder	Head features	Measurement and preprocessing	Statistical / ML method	Performance and limitations
Boutaleb et al. 2025 ²²	32 patients with severe depression; psychiatrist-rated psychological anxiety	Pose angular displacement, speed, acceleration ; moving/steady segmentation features; transition rates	Videos of informal interviews from CALYPSO dataset; segmentation-driven 283-feature summary; stable feature selection	Lasso regression with sequential feature selection and 10-fold CV	Best model: MAE 0.31, R² 0.87, accuracy 0.75 with 14 features. Without segmentation: MAE 0.90, R² 0.10 . Limitation: anxiety assessed only inside severe-depression cohort.
Stratou et al. 2013 ³⁷	53 virtual-human interview videos; depression and PTSD	Interpretable indicators from affect, expression variability, motor variability ; accessible source highlights head-rotation variation for PTSD	Semi-structured virtual-human interviews; full tracker details not exposed in accessible sources	Automatic assessment with gender-aware analysis	Key contribution is conceptual: gender matters, and head-rotation variation appears relevant for PTSD. Limitation: accessible sources do not expose complete pipeline or accuracy metrics.

Study	Population and disorder	Head features	Measurement and preprocessing	Statistical / ML method	Performance and limitations
AnxietyFaceTrack 2025 preprint ³⁸	91 participants in unstaged social setting; social-anxiety screening	Head location and rotation ; pose_Rx repeatedly important	OpenFace , smartphone video at 30 fps ; 714 features; feature-importance filtering	Random Forest , 5-fold CV	Head-pose-only combined features about 85% multiclass accuracy; top-20%-feature model 91% multiclass. Limitation: preprint; questionnaire/state-based labels, not clinician-confirmed disorder.

Several cross-study patterns emerge from that matrix. First, **depression** is the only area where head movement has been shown repeatedly in **clinically characterized interview data**, and the convergent signal is **psychomotor slowing / reduced behavioral variability** rather than one single “PTSD-style head pose” or “anxiety-style head pose.” Second, the strongest head features are usually **dynamic** ones—velocity, acceleration, bout structure, transitions, segmented-state statistics—not raw pose means. Third, **simple totals** can be too blunt: Kayıř’s coarse total-and-rapid head metrics were null, whereas Boutaleb’s segmented temporal descriptors were highly predictive. Fourth, the best-performing studies often use head pose **as one stream inside a richer multimodal model**, not as a stand-alone diagnostic tool. ³⁹

Gaps, reproducibility, confounders, ethics, and recommendations

The biggest scientific gap is disorder imbalance. The field has meaningful **MDD head-movement evidence**, but **PTSD** and **anxiety-disorder** evidence is still either multimodal, indirect, or clinically thin. Even for depression, the literature is fragmented across **different outcome definitions**: DSM case-control status, HRSD remission/depression bands, BDI thresholds, self-report states, or symptom-dimension regression. The interpretation paper explicitly notes that BlackDog, Pitt, and AVEC used different depression assessment tools and procedures, making direct score harmonization difficult; it also notes that environmental and protocol differences likely influenced generalization. ⁴⁰

Reproducibility remains a serious issue. Some studies rely on **controlled clinical rooms, fixed seating, specific camera angles**, and **manual or semi-manual quality control**; others analyze long challenge-corpus videos with different elicitation tasks; still others use smartphone or POV-glasses recordings. As a result, “head movement” does not mean the same thing operationally across papers. One study may mean pitch/yaw/roll angle derivatives with robust smoothing; another may mean percentile speed from an RGB-D skeleton; another may mean transitions between moving and steady states; another may mean kineme frequencies. Those are related signals, but they are not interchangeable biomarkers. ⁴¹

Several confounders recur or remain under-measured. **Medication and treatment status** are one: Dibeklioglu's participants came from a treatment trial, some randomized to **SSRI** and some to **Interpersonal Psychotherapy**, which could plausibly affect psychomotor output even though treatment arm was not the focus of the paper. **Comorbidity** is another: Dibeklioglu explicitly notes the value of diagnostic interview in ruling out related disorders such as PTSD, whereas the anxiety-severity CALYPSO study is nested inside a severe-depression cohort, so anxiety-specific effects are inseparable from depressive context. **Age and sex** also vary substantially across studies, from young student screening samples to adult clinical cohorts; some papers match demographics carefully, others do not. **Culture and language** likely matter as well: Bhatia's synchrony paper explicitly lists language, dialect, culture, and ethnicity as uncontrolled influences on synchrony, and the Alghowinem interpretation paper frames cross-dataset generalization as a major challenge. ⁴²

Ethically, the reviewed systems work with **identifiable face video, and often synchronized audio**, captured in unusually sensitive contexts: trauma interviews after emergency care, psychiatric diagnostic interviews, longitudinal treatment sessions, and clinician-worn POV recordings. That creates privacy risks far beyond ordinary computer-vision datasets. It also raises fairness and governance questions, because small psychiatric datasets are easy to overfit and difficult to audit externally. In my view, the responsible near-term framing is **clinical decision support**, not automated diagnosis. The safest deployment model is to extract **pose trajectories and derived features as early as possible**, retain raw media only under explicit consent and short retention rules, and preserve a human clinician as the final decision-maker. ⁴³

The methodological improvements the field now needs are concrete. Future studies should use **pre-registered primary head features**; report **camera geometry, resolution, frame rate, tracker confidence thresholds, smoothing method, derivative formulas, missing-frame policy, and calibration procedure**; and separate **head-only, multimodal, and incremental-gain** analyses so that the unique value of head movement can actually be computed. At minimum, each paper should publish a compact protocol describing: how pitch/yaw/roll were derived; whether values are in degrees or radians; whether they are camera-relative or face-relative; whether derivatives were raw or smoothed; whether temporal windows overlapped; and whether statistics were computed per frame, per turn, per answer, or per whole interview. Those details are missing or inconsistent in much of the literature reviewed here. ⁴⁴

For study design, the strongest next step would be a **multi-site prospective cohort** that uses the same protocol across **MDD, PTSD, GAD/SAD**, and healthy controls, with both categorical diagnosis and dimensional scales such as **HAM-D/HRSD, PCL-5 or CAPS-5**, and **GAD-7 / SIAS / SPIN**. Each session should include a short calibration clip with deliberate nods, shakes, neutral fixation, and free conversation; then both a structured symptom interview and a more naturalistic social segment. Collecting a **depth or stereo subset** for calibration error estimation would allow monocular systems to be benchmarked without making depth hardware mandatory at scale. A **telehealth subset** should be included from the outset, because one of the most attractive use cases for these markers is remote monitoring. The reviewed papers strongly suggest that behavior changes with interview context, so context should be treated as a **designed factor**, not as nuisance variation. ⁴⁵

My bottom-line assessment is this: **head movement is already useful enough to be included in digital phenotyping stacks for depression, probably useful enough to investigate seriously for PTSD and anxiety, and not yet mature enough to be marketed as a stand-alone psychiatric biomarker**. The most defensible current claim is that head movement provides an **interpretable, low-burden, behaviorally meaningful signal of psychomotor state and social responsiveness**, especially when modeled

dynamically and fused with facial, vocal, linguistic, or physiologic channels. The most important future job is not to invent yet another classifier, but to make head-movement measurement **standardized, reproducible, externally validated, and clinically interpretable.** 46

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